

**Statement of Objective (SOO)/Technical Requirements Package (TRP)**

**USAF CIL NIST-Traceable radioactive standards**

**9 Feb 2026**

## **1. SOO/TRP for Equipment**

USAF CIL NIST-Traceable radioactive standards

## **2. Purpose**

The purpose of this effort is to procure certified NIST-traceable radioactive reference standards prepared in specific geometries/matrices supplied by Radiochemistry Laboratory's (CIL). Specific geometries are detailed in section 3.

## **3. Technical Requirements**

### **3.1 List of required standards and geometries/matrices**

1. Product Code: 8401-GF-RASA – 2.00 Each  
8401

148.00 kBq ( 4.00 uCi )

Activity Tolerance: +20% / -20%

Cd-109, Co-57, Ce-139, Hg-203, Sn-113, Cs-137, Y-88, Co-60, Am-241

Composite of six 5in x 20in RASA filters in heat sealed plastic bag, dimensions of active area 4.25in x 16in, active material to be deposited on 5in x 20in 0.5mg/cm<sup>2</sup> Mylar insert in between the 3rd and 4th RASA filters

(filters to be inserted back into original white plastic covering prior to being deflated and heat sealed in a 4mil poly sleeve) Calibrated, NIST Traceable

2. Product Code: 8403-EG-SDLD – 1.00 Each  
8403

77.70 kBq ( 2.10 uCi )

Activity Tolerance: +20% / -20%

Cd-109, Co-57, Ce-139, Hg-203, Sn-113,

Cs-137, Y-88, Co-60, Am-241, Pb-210

0.6 g/cc +/-10% polyester matrix

Fill Volume: 26 mL

Customer Supplied 53 mm parkway PPI petri (GFU)

Filled w/ silicone adhesive to prevent solid fill from dislodging

Calibrated, NIST Traceable

3. Product Code: 8403-GF-FP – 1.00 Each  
8403

103.60 kBq ( 2.80 uCi )

Activity Tolerance: +20% / -20%

Cd-109, Co-57, Ce-139, Hg-203, Sn-113,

Cs-137, Y-88, Co-60, Am-241, Pb-210

Customer supplied 24mm dia. paper filter in tape

Active Diameter: 20 mm

mounted onto Customer Supplied 1.75 in stainless steel disk  
activity facing away from the disk  
Calibrated, NIST Traceable

4. PRODUCT CODE 8403-GF-FP - 1.00 each

8403

148.00 kBq ( 4.00 uCi )

Activity Tolerance: +20% / -20%

Cd-109, Co-57, Ce-139, Hg-203, Sn-113,

Cs-137, Y-88, Co-60, Am-241, Pb-210

Customer Supplied 16 in dia. U1 filter in poly bag

Active Diameter: 15.5 in, 0.5mg/cm<sup>2</sup> Mylar foil cover

Folded into 1/16 fold geometry (folded filter to be pressed flat,

bag to be deflated prior to heat sealing in a 4mil poly sleeve) Calibrated, NIST Traceable

5. Product Code: 8409-EG-SD - 1.00 each

8409

185.00 kBq ( 5.00 uCi )

Activity Tolerance: +20% / -20%

Cd-109, Co-57, Ce-139, Hg-203, Sn-113, Cs-137,

Y-88, Co-60, Am-241, Pb-210, Mn-54, Zn-65

1.17g/cc +/-3% polyester matrix, NIST traceable

Fill Volume: 10mL

Wheaton HDPE LSV with silicone adhesive to prevent solid fill from dislodging, Calibrated,  
NIST Traceable

6. Product Code: 8409-EG-SD - 1.00 each

8409

37.00 kBq ( 1.00 uCi )

Activity Tolerance: +20% / -20%

Cd-109, Co-57, Ce-139, Hg-203, Sn-113, Cs-137,

Y-88, Co-60, Am-241, Pb-210, Mn-54, Zn-65

1.17g/cc +/-3% polyester matrix, NIST traceable

Fill Volume: 10mL

Wheaton HDPE LSV with silicone adhesive to prevent solid fill from  
dislodging

Calibrated, NIST Traceable

1. PRODUCT CODE: 8405-EG-SD – 1.00 Each

8405

37.00 kBq ( 1.00 uCi )

Activity Tolerance: +20% / -20%

Cd-109, Co-57, Ce-139, Hg-203, Sn-113, Cs-137, Y-88, Co-60, Am-241, Mn-54, Zn-65

1.17g/cc +/-3% polyester matrix, NIST traceable Fill Volume: 10mL

Wheaton HDPE LSV w/ silicone adhesive to prevent solid fill from dislodging Calibrated,  
NIST Traceable

### **Warranty and Support characteristics**

- Nuclear standards shall be accompanied by certified NIST-traceable documentation.

## **4. Technical Contractual Requirements**

### **4.1 Remanufactured/Refurbished Products**

Remanufactured/Refurbished products shall not be accepted.

## **5. Delivery NLT 30 Sep 2026**